

BREATHEBOX · PROBLEM STATEMENT

BreatheBox problem statement

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DOCUMENT CONTROL

Document ID	BBX-PRB-001	Version	0.5
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breathebox		
Source	docs/01-problem.md @ d61ee78		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget approved by Amish; constraint now \$255
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update; sash-first window type and budget position per BBX-DDR-001 (adopted for TRL 3, open for Amish's review); open questions updated
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (users, context, constraints, prior work, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

BreatheBox problem statement

Most homes have no mechanical ventilation, so the people in them choose between stale, damp air and an open window that throws away heat or cooling. Whole-house heat recovery ventilators solve this in new, airtight buildings, but they need ducts through ceilings and walls, which renters cannot install and owners of older homes rarely pay for. There is no open, low-cost, renter-installable way to ventilate one room with heat recovery.

■ Why it matters

People spend most of their lives indoors. The US Environmental Protection Agency notes that Americans spend about 90 % of their time indoors and that indoor concentrations of some pollutants are often 2 to 5 times higher than typical outdoor levels ([US EPA, Report on the Environment: Indoor Air Quality](#)). Bedrooms are the worst case: two people sleep for eight hours behind a closed door and a closed window.

Energy is the other half of the problem. In the European Union, buildings use about 40 % of energy, about 80 % of the energy used in homes goes to heating, cooling and hot water, and 85 % of buildings were built before 2000 ([European Commission, Energy Performance of Buildings Directive](#)). As these homes are sealed and insulated, air leakage falls and moisture and CO₂ build up unless ventilation is added. In England, 7 % of social rented homes had a damp problem in 2023, and the death of two-year-old Awaab Ishak in 2020 after prolonged exposure to mould led to a law that sets deadlines for landlords to fix damp and mould hazards ([UK Government, Awaab's Law](#)).

Outdoor air is not always clean either. The World Health Organization reports that 99 % of the world's population lived in places where its air quality guideline levels were not met in 2019 ([WHO, Ambient air quality and health](#)). An open window brings that air in unfiltered; a ventilator can filter it.

■ Users and context

User	Situation	What they need
Renters in older flats and houses	Cannot drill, duct or change the building; often have one problem room (a bedroom or a small flat's main room)	A unit that goes into an existing window and comes out again without damage
Owners of older, poorly ventilated homes	Condensation and mould on cold walls; high heating bills; no budget for a whole-house system	Continuous fresh air to one or two rooms at low running cost
Households in polluted cities	Windows kept shut against traffic or smoke; stuffy bedrooms	Filtered supply air and a visible CO ₂ reading
Social housing providers and energy retrofit programs	Must fix damp and mould quickly across many homes	A low-cost, repeatable, documented measure that installs in minutes
Makers and community repair groups	Want to build, adapt and fix the unit locally	Open drawings, common parts and no special tools

Operating context. One room of 20 to 40 m³ (for example, a 3 x 4 m bedroom with a 2.5 m ceiling holds 30 m³), one or two occupants, a window with a sill 800 to 1,000 mm above the floor, and a standard wall socket nearby. Outdoor air from about -10 °C to 35 °C. The first target window is the vertical sliding sash (single- or double-hung) common in North America and the United Kingdom, because the lower sash can close down onto an insert panel. This sash-first choice was decided by Amish on 2026-09-25 (BBX-DDR-001 item A2; BBX-DDR-002).

■ Constraints

- Garage-buildable prototype, \$255 USD in parts (`project.yaml` ; raised from \$220 to \$250 by Amish on 2026-09-25 and to \$255 on 2026-09-26, BBX-DDR-002). The priced BOM is \$255 (BBX-CAL-001).
- No drilling of the window frame or wall; installs and removes without damage, so renters can use it.
- Low voltage only inside the unit (24 V SELV from a certified plug-in adapter); no mains wiring by the builder.
- Common, replaceable parts: standard fans, filter media cut to size, a spare-part HRV core.
- Quiet enough for a bedroom at night.
- Open licenses: CERN-OHL-S-2.0 for hardware, MIT for firmware and scripts.

■ Out of scope

- Whole-house ventilation, ducted systems and replacement of building code ventilation.
- Heating or cooling the room (the unit recovers heat; it does not add it beyond fan heat).
- Kitchen and bathroom extraction, which need higher extract rates, and removal of combustion products from stoves or open-flued appliances.
- Medical or health claims. The CO₂ display is an indicator of ventilation, not a health measurement.

■ Prior work

- **Decentralized single-room ventilators.** Commercial through-wall units exist, for example the LUNOS e² family of small decentralized fans with heat recovery ([LUNOS, heat recovery products](#)). The LUNOS page, checked on 2026-09-25, describes the e² as one of the smallest decentralized fans but does not state how it is installed. Through-wall units of this kind need an opening in the wall, which renters generally cannot make.
- **Plate heat exchangers.** Fixed-plate exchangers reach 70 to 90 % sensible efficiency at low pressure drop, but have a high chance of frosting in cold climates ([Wikipedia, Heat recovery ventilation](#)). BreatheBox uses a plate counterflow core and needs a frost strategy.
- **Window air conditioners and window fans** show that a sash window with a filler panel is a familiar, renter-friendly mounting. They move air or heat but do not recover heat from exhaust air.
- **CO₂ as a ventilation indicator.** Health Canada's residential guideline sets a long-term exposure limit of 1,000 ppm CO₂ (24-hour average) and notes that ventilation is the primary means of removing CO₂ indoors ([Health Canada, Residential indoor air quality guidelines: carbon dioxide](#)).

We have not found an open-source, window-mounted, counterflow heat recovery ventilator with published design files. This is a finding from a limited search and should be checked at TRL 3.

■ Open questions

- [] Which windows dominate in the first user group? The first version targets sash windows (BBX-DDR-001 item A2); casement and tilt-and-turn windows (common in continental Europe) need a later insert.
- [] Is a 260 mm raised sash acceptable for security, or is a locking bar needed as standard?
- [] What noise level do users accept at night, and at what airflow? BBX-CAL-001 estimates about 39 dB(A) at 1 m at 50 m³/h and 30 dB(A) only near 32 m³/h, A quiet night mode at 32 m³/h (about 30 dB(A), BBX-DDR-002) is now part of the design, so the open part is whether users accept it and its overnight CO₂ of about 1,300 ppm.
- [] How cold does it get where the first users live, and how often will the frost strategy run?

- [] Who are the first co-design partners (a tenants' group, a social landlord or a retrofit program)? Proposed, awaiting Amish (BBX-DDR-001 item O1).